

# Chamber findings — day 3, top 5 setups

2026-09-25 (+ rerun 2026-09-30) · 95 tones 257 Hz–6.35 kHz, 3–5 kHz omitted · 11 calibrated capsules · sphere on the axis, arc vertical

**Coefficient** = room error below 3 kHz: rms over all tones and capsules of each capsule's level relative to the arc mean, dB, lower is flatter. Repeat floor 0.01 dB, handling ~0.04 dB, so **the top four are tied** (span 0.014 dB). The last config is 0.086 dB behind: a real difference. Today's rerun of the last config scored 1.399 (+0.013): consistent with the chamber still being in that state at 15:48.

| SETUP                                                                                                                                                                       | ROOM ERROR<br><3 KHZ | VS BEST | 250-400 | 400-630 | 630-1K | 1-1.6K | 1.6-3K | QUALIFIED<br>FROM* | BANDS<br>FAILED* |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|---------|---------|---------|--------|--------|--------|--------------------|------------------|
| <b>last config</b><br>2026-09-25/carpet-reordered                                                                                                                           | <b>1.387</b>         | +0.086  | 1.99    | 1.52    | 1.47   | 0.85   | 0.96   | 2000 Hz            | 4                |
| Last configuration of day 3 (19:20): the full carpet on standoffs, layer order changed (bottom layer moved to the top).                                                     |                      |         |         |         |        |        |        |                    |                  |
| <b>cleanup-1</b><br>2026-09-25/cleanup-1                                                                                                                                    | <b>1.301</b>         | –       | 1.56    | 1.62    | 1.43   | 0.81   | 1.03   | 1000 Hz*           | 3                |
| BEST. The accepted state (speaker backed off 5 cm from the ring plane, raised thick carpet) with the loose felt sheets removed.                                             |                      |         |         |         |        |        |        |                    |                  |
| <b>backed-5cm</b><br>2026-09-25/backed-5cm                                                                                                                                  | <b>1.303</b>         | +0.002  | 1.57    | 1.61    | 1.44   | 0.81   | 1.03   | 1000 Hz*           | 3                |
| The accepted state: speaker backed off 5 cm from the ring plane, raised thick carpet.                                                                                       |                      |         |         |         |        |        |        |                    |                  |
| <b>closer</b><br>2026-09-25/closer-a                                                                                                                                        | <b>1.311</b>         | +0.011  | 1.99    | 1.28    | 1.29   | 0.86   | 0.97   | 315 Hz*            | 1                |
| Speaker moved ~20 cm closer to the ring, still on the axis; morning floor stack of day 3 (after the blue absorbing carpet was added). Repeat `closer-b` agrees to 0.006 dB. |                      |         |         |         |        |        |        |                    |                  |
| <b>cleanup-2</b><br>2026-09-25/cleanup-2                                                                                                                                    | <b>1.314</b>         | +0.014  | 1.61    | 1.72    | 1.40   | 0.79   | 0.98   | 500 Hz             | 2                |
| The accepted state after more junk was removed (what, not stated).                                                                                                          |                      |         |         |         |        |        |        |                    |                  |

- \*ISO 3745 analogue on band levels, section 2. "Qualified from" = the lowest one-third-octave band from which no band up to 2.5 kHz is clearly over the limit (the way METU and others state a cut-off); \* = that band is marginal. "Bands failed" is out of 11.
- The single coefficient ranks cleanup-1 first, but it hides *where* the error is. Bands clearly over the limit, of 11: last config 4, cleanup-1 3, backed-5cm 3, closer 1, cleanup-2 2. Closer and cleanup-2 are the cleanest by that count although they score 0.01 dB behind.
  - Worst band: 250–400 Hz: last config, closer; 400–630 Hz: cleanup-1, backed-5cm, cleanup-2. Best band: 1000–1600 Hz: last config, cleanup-1, backed-5cm, closer, cleanup-2.
  - Different floors: closer was measured on the morning stack of day 3, the rest on later stacks. Across days the map drifts 0.1–0.2 dB even when the score does not.

2. Which frequencies to avoid

**Rule, taken from how chambers are qualified.** ISO 3745 compares the level at each one-third-octave band with the ideal and allows ±1.5 dB up to 630 Hz and ±1.0 dB from 800 Hz to 5 kHz (the same table in six independent papers). A chamber is then reported as qualified between two bands, with the lowest one its cut-off. Nash 2019 found a chamber that passed with random noise and failed with pure tones, so both are reported here.

**What we changed.** ISO compares a microphone traverse with the inverse-square law. We have eleven fixed positions and an axisymmetric source, so the reference is the arc mean. Two tests per band: **band level** (each capsule's mean over the ~8 tones in the band; worst capsule against the limit) and **pure tones** (share of individual tone×capsule cells inside the limit). Band verdicts within ±0.09 dB of the limit are marked marginal: that is how far the same statistic moved between the last config and its rerun.

| Band level: worst capsule vs arc mean (dB)                                  |             |           |            |        |           | Pure tones: % of tonexcapsule cells |           |            |        |           |
|-----------------------------------------------------------------------------|-------------|-----------|------------|--------|-----------|-------------------------------------|-----------|------------|--------|-----------|
| orange = over the ISO 3745 limit, yellow = within ±0.09 of inside the limit |             |           |            |        |           |                                     |           |            |        |           |
| 250 Hz ±1.5 (3 tones)                                                       | 1.88        | 1.41      | 1.34       | 2.38   | 0.84      | 52                                  | 88        | 88         | 61     | 97        |
| 315 Hz ±1.5 (8 tones)                                                       | 0.89        | 1.76      | 1.77       | 1.51   | 1.99      | 56                                  | 62        | 61         | 62     | 58        |
| 400 Hz ±1.5 (8 tones)                                                       | 1.41        | 2.36      | 2.33       | 1.50   | 2.36      | 59                                  | 62        | 61         | 68     | 59        |
| 500 Hz ±1.5 (8 tones)                                                       | 1.99        | 1.07      | 1.07       | 0.91   | 1.15      | 59                                  | 67        | 66         | 67     | 65        |
| 630 Hz ±1.5 (8 tones)                                                       | 1.49        | 0.97      | 0.97       | 0.98   | 0.65      | 61                                  | 62        | 65         | 77     | 60        |
| 800 Hz ±1 (8 tones)                                                         | 1.12        | 1.15      | 1.16       | 0.70   | 0.80      | 55                                  | 47        | 47         | 47     | 50        |
| 1000 Hz ±1 (9 tones)                                                        | 0.57        | 0.95      | 0.95       | 0.55   | 0.95      | 73                                  | 69        | 68         | 80     | 69        |
| 1250 Hz ±1 (8 tones)                                                        | 1.02        | 0.96      | 0.97       | 0.68   | 0.73      | 73                                  | 83        | 83         | 80     | 82        |
| 1600 Hz ±1 (8 tones)                                                        | 1.15        | 0.61      | 0.61       | 0.62   | 0.56      | 76                                  | 76        | 77         | 68     | 81        |
| 2000 Hz ±1 (8 tones)                                                        | 0.77        | 0.61      | 0.62       | 0.69   | 0.71      | 82                                  | 77        | 80         | 74     | 77        |
| 2500 Hz ±1 (8 tones)                                                        | 0.75        | 0.66      | 0.67       | 0.93   | 0.71      | 65                                  | 67        | 67         | 62     | 68        |
|                                                                             | last config | cleanup-1 | backed-5cm | closer | cleanup-2 | last config                         | cleanup-1 | backed-5cm | closer | cleanup-2 |

| BAND    | LIMIT DB | TONES | SETUPS OVER THE LIMIT (BAND LEVEL), OF 5 | PURE TONES IN LIMIT, MEDIAN | VERDICT         |
|---------|----------|-------|------------------------------------------|-----------------------------|-----------------|
| 250 Hz  | ±1.5     | 3     | 2 clear + 1 marginal                     | 88 %                        | setup-dependent |
| 315 Hz  | ±1.5     | 8     | 3 clear + 1 marginal                     | 61 %                        | avoid           |
| 400 Hz  | ±1.5     | 8     | 3 clear + 1 marginal                     | 61 %                        | avoid           |
| 500 Hz  | ±1.5     | 8     | 1 clear + 0 marginal                     | 66 %                        | setup-dependent |
| 630 Hz  | ±1.5     | 8     | 0 clear + 1 marginal                     | 62 %                        | borderline      |
| 800 Hz  | ±1       | 8     | 3 clear + 0 marginal                     | 47 %                        | avoid           |
| 1000 Hz | ±1       | 9     | 0 clear + 3 marginal                     | 69 %                        | borderline      |
| 1250 Hz | ±1       | 8     | 0 clear + 3 marginal                     | 82 %                        | borderline      |
| 1600 Hz | ±1       | 8     | 1 clear + 0 marginal                     | 76 %                        | setup-dependent |
| 2000 Hz | ±1       | 8     | 0 clear + 0 marginal                     | 77 %                        | holds           |
| 2500 Hz | ±1       | 8     | 0 clear + 1 marginal                     | 67 %                        | borderline      |

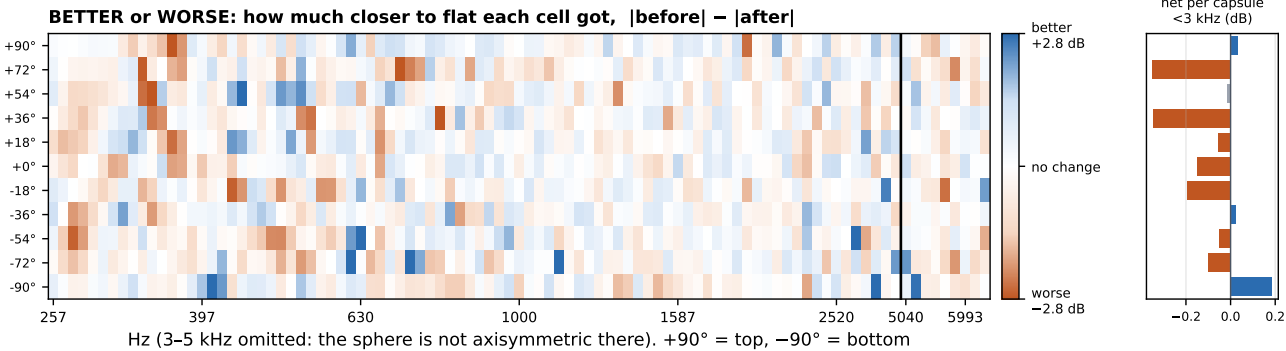
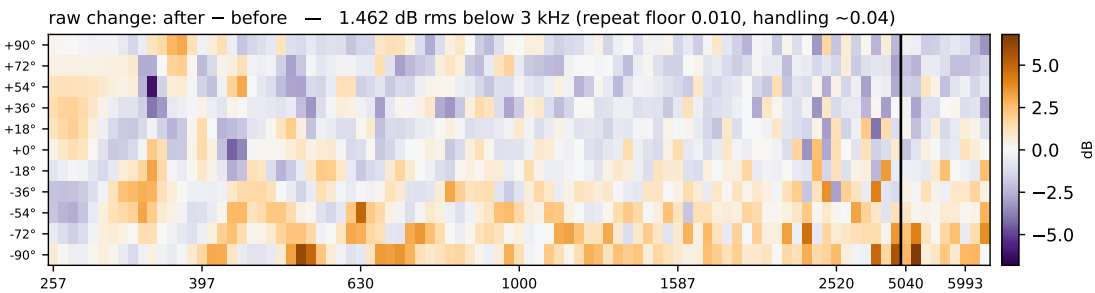
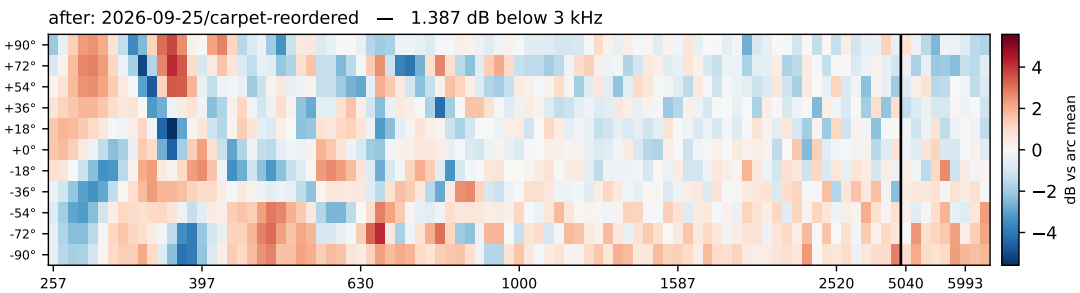
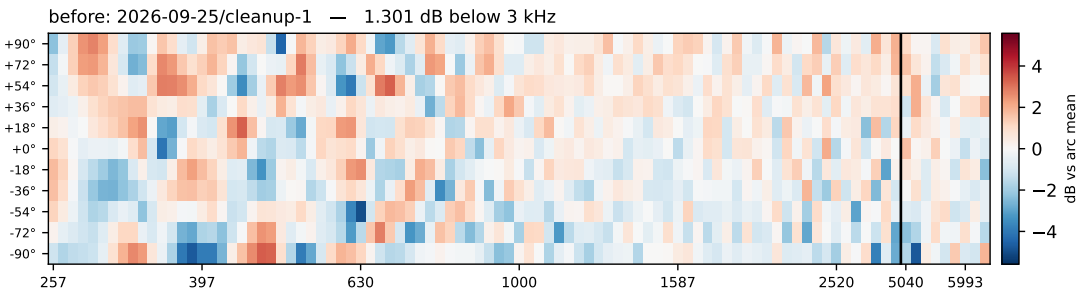
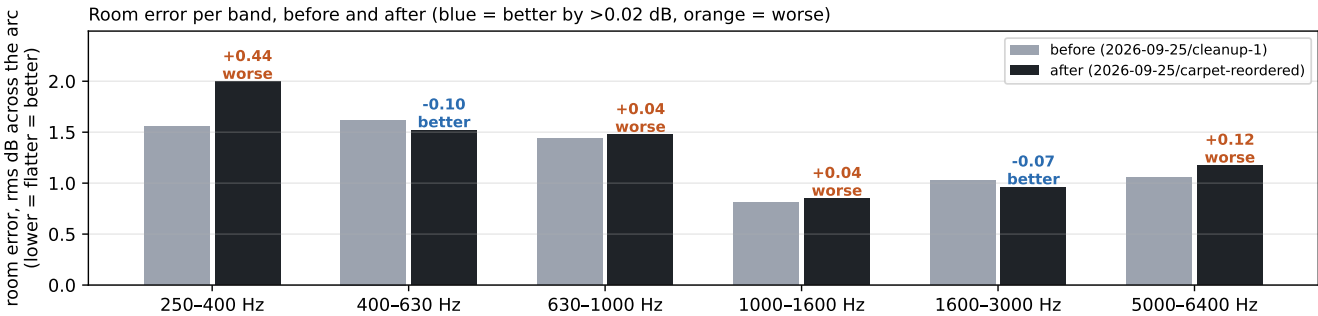
avoid = clearly over the limit in 3 or more of the 5 setups; marginal cases are last config at 630 Hz, last config at 1250 Hz, cleanup-1 at 250 Hz, cleanup-1 at 1000 Hz, cleanup-1 at 1250 Hz, backed-5cm at 1000 Hz, backed-5cm at 1250 Hz, closer at 315 Hz, closer at 400 Hz, closer at 2500 Hz, cleanup-2 at 1000 Hz.  
setup-dependent = 1–2; borderline = none clearly over but some marginal; holds = neither. These cut-offs are ours, not from a standard.

- **Avoid:** 315, 400, 800 Hz. **Clean in all five:** 2000 Hz. **Borderline** (no clear failure, but at least one setup within the scatter of the limit): 630, 1000, 1250, 2500 Hz. **Depends on the setup:** 250, 500, 1600 Hz (250 Hz fails only in last config, closer; 500 Hz fails only in last config; 1600 Hz fails only in last config).
- **Pure tones are not qualified anywhere.** No band has all cells inside the limit; the median is 47–82 %, lowest at 800 Hz.
- **Never usable:** 3–5 kHz (the sphere's rocking mode, a source fault). **Indicative only:** 5–6.4 kHz. **Not measured:** below 257 Hz, where blade tones sat at 216 and 238 Hz (and 258 Hz) in the September runs, and the 250 Hz band has only 3 tones.

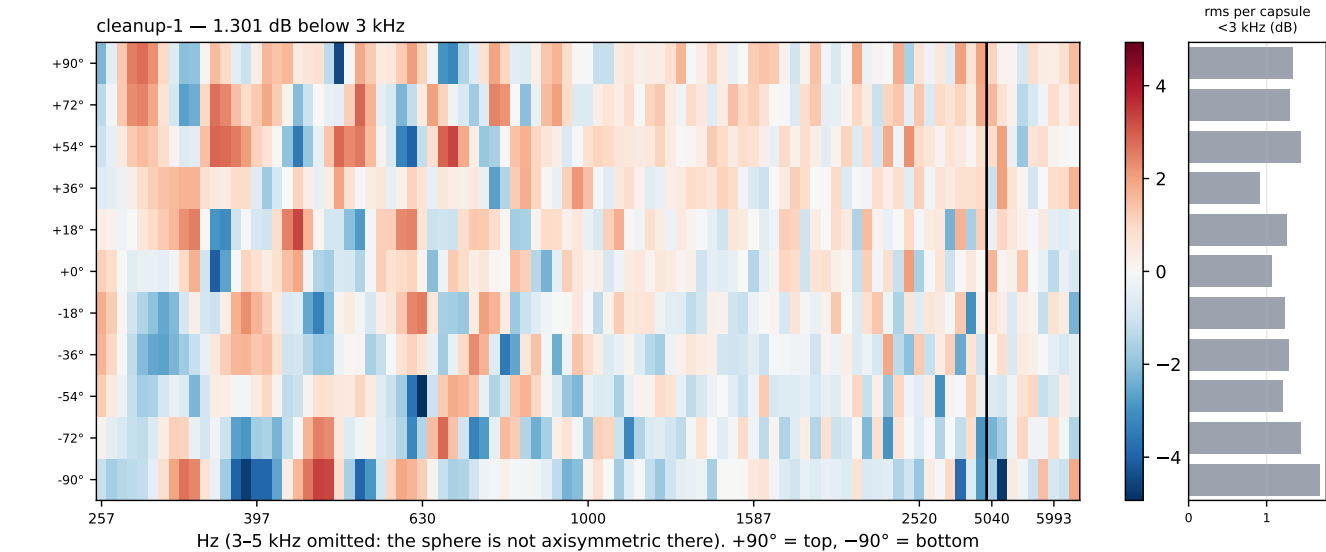
**Limits of this rule.** Deviation from the arc mean cannot see an error shared by all eleven positions, such as a uniform reflection. Band verdicts hinge on the single worst of eleven capsules. The tolerance table belongs to a traverse test we never ran; treat the verdicts as a ranking of bands, not a certificate.

last config vs best (cleanup-1): room error below 3 kHz 1.301 → 1.387 dB (+0.086, worse)

2026-09-25/cleanup-1 → 2026-09-25/carpet-reordered · below 3 kHz, 427 of 946 cells moved >0.05 dB toward flat and 481 away from it · 95 tones · 11 calibrated capsules

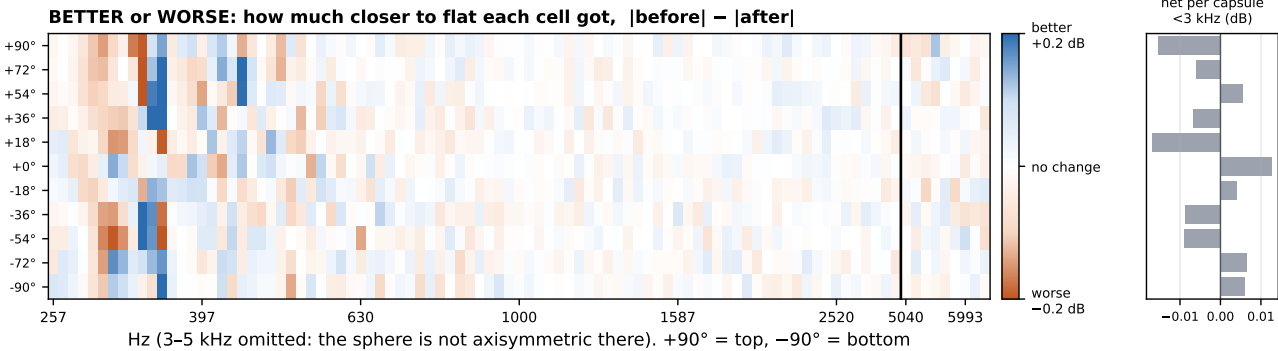
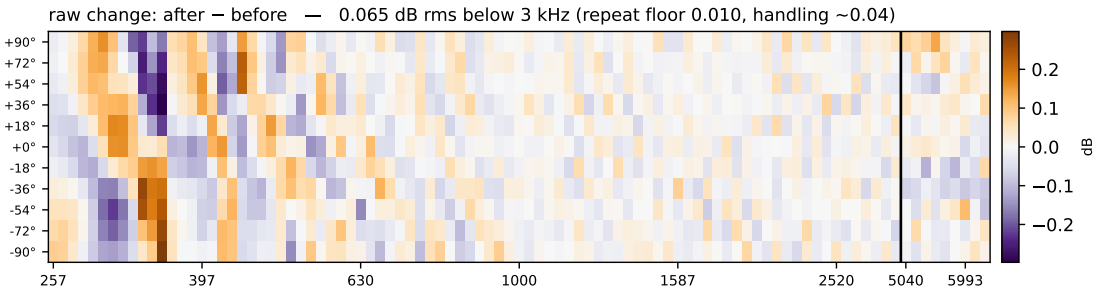
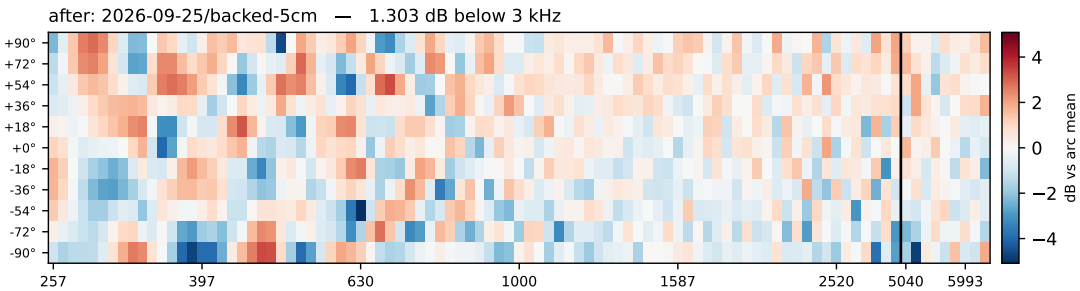
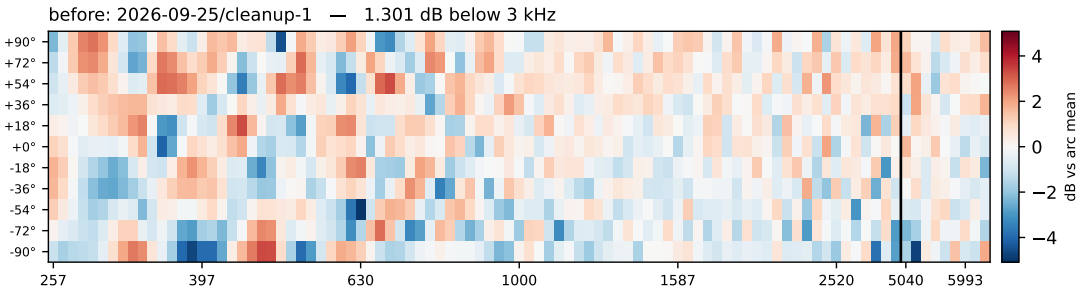
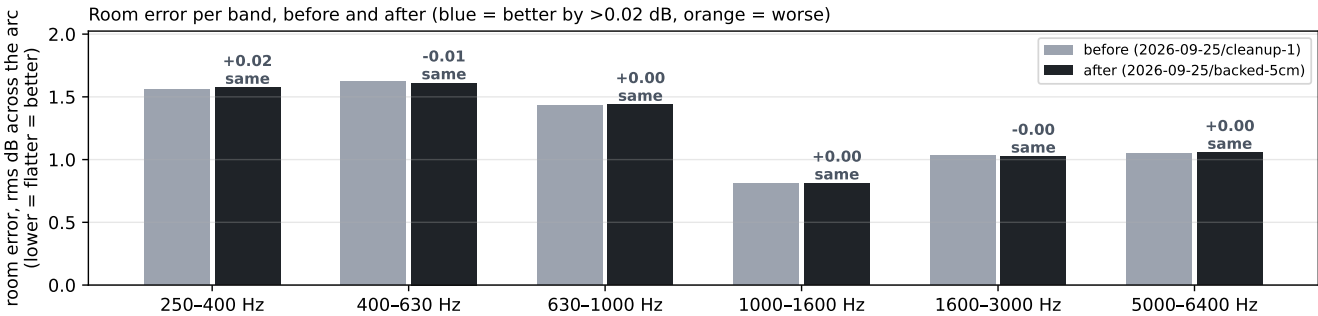


2026-09-25/cleanup-1 · 95 tones · 11 calibrated capsules · the four pages that follow it are drawn against this map



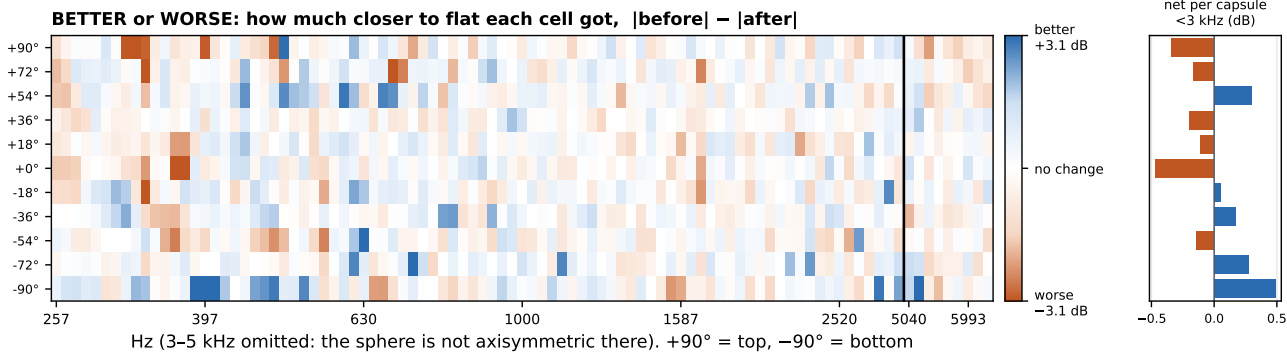
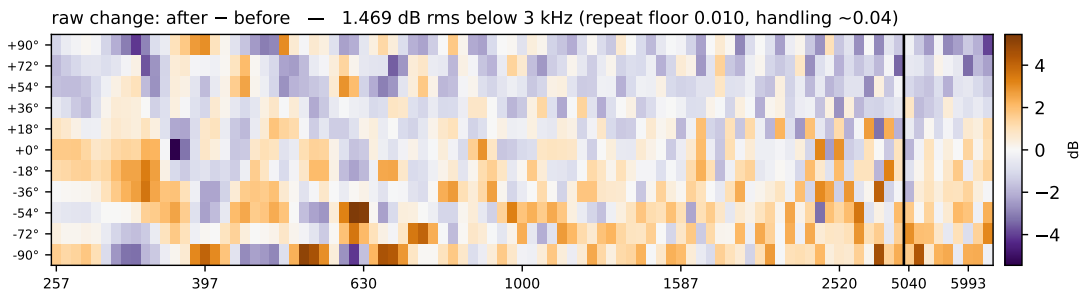
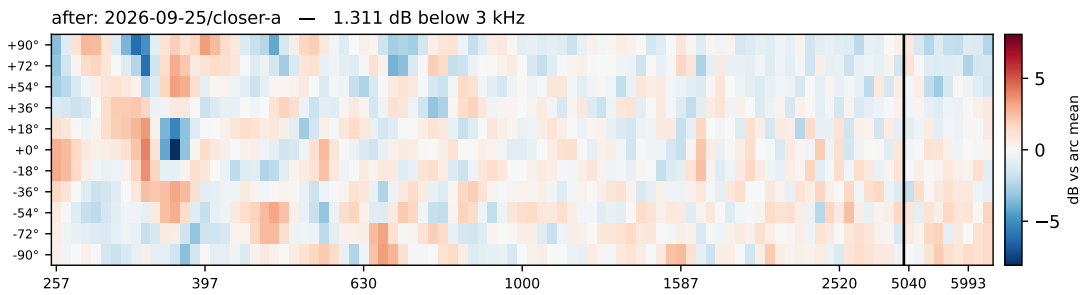
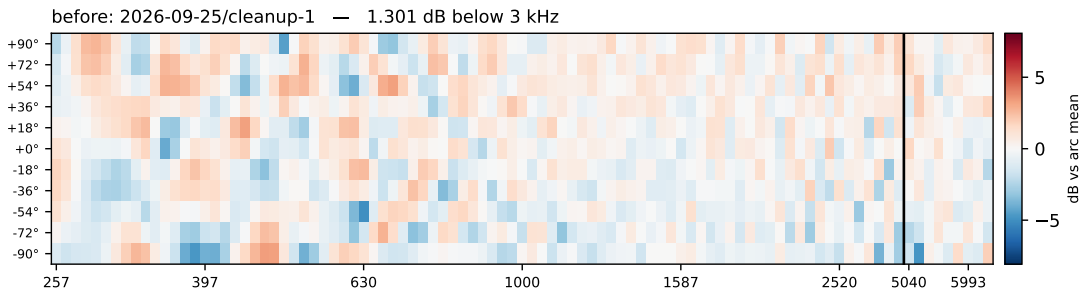
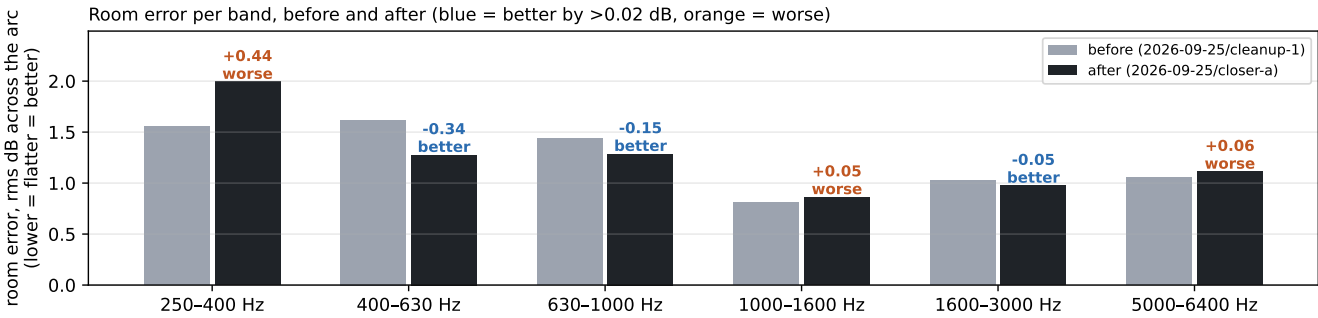
backed-5cm vs best (cleanup-1): room error below 3 kHz 1.301 → 1.303 dB (+0.002, same)

2026-09-25/cleanup-1 → 2026-09-25/backed-5cm · below 3 kHz, 143 of 946 cells moved >0.05 dB toward flat and 149 away from it · 95 tones · 11 calibrated capsules



closer vs best (cleanup-1): room error below 3 kHz 1.301 → 1.311 dB (+0.011, worse)

2026-09-25/cleanup-1 → 2026-09-25/closer-a · below 3 kHz, 464 of 946 cells moved >0.05 dB toward flat and 434 away from it · 95 tones · 11 calibrated capsules



cleanup-2 vs best (cleanup-1): room error below 3 kHz 1.301 → 1.314 dB (+0.014, worse)

2026-09-25/cleanup-1 → 2026-09-25/cleanup-2 · below 3 kHz, 398 of 946 cells moved >0.05 dB toward flat and 392 away from it · 95 tones · 11 calibrated capsules

